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FINAL PROJECT

BSIT INDUSTRY-READY PORTFOLIO

(Student Guidelines and Step-by-Step Template)

"My BSIT Industry-Ready Professional Portfolio"

Project Purpose

This project is designed to help graduating BSIT students demonstrate the knowledge, technical skills, projects, problem-solving abilities, and professional competencies they have acquired throughout the BSIT program.

The final portfolio should demonstrate:

WHAT I KNOW + WHAT I CAN DO + WHAT I HAVE BUILT + HOW I SOLVE PROBLEMS + HOW I CAN PRESENT MYSELF AS AN IT PROFESSIONAL

I. PROJECT OBJECTIVES

1. Create a professional personal portfolio website.
2. Present technical competencies in Information Technology.
3. Demonstrate programming skills through actual projects.
4. Demonstrate database and SQL skills.
5. Demonstrate web or application development skills.
6. Demonstrate basic networking and troubleshooting skills.
7. Demonstrate cybersecurity awareness.
8. Demonstrate data-analysis skills.
9. Demonstrate basic cloud/deployment knowledge.
10. Demonstrate responsible use of AI tools.
11. Organize and document software projects professionally.
12. Use Git and GitHub for version control.
13. Create a professional resume.
14. Prepare a professional LinkedIn profile.
15. Demonstrate technical communication skills.

16. Participate in a mock IT job interview.
17. Defend and explain a completed IT project.
18. Develop a personal career-development plan.

II. FINAL PORTFOLIO STRUCTURE

Your portfolio website must contain the following major sections:

- HOME
- 1. PROFESSIONAL PROFILE
- 2. TECHNICAL SKILLS
- 3. PROJECTS
- 4. PROBLEM-SOLVING
- 5. EMPLOYABILITY
- 6. RESUME
- 7. CERTIFICATES / ACHIEVEMENTS
- 8. GITHUB
- 9. CONTACT
- 10. REFLECTION

STEP 1 — Define Your IT Professional Identity

Before creating the website, answer the following:

Full Name: _____
BSIT Specialization/Area: _____
Target IT Position: _____

Preferred IT Field:

☐ Web Development
☐ Software Development
☐ Mobile Development
☐ Database Administration
☐ Data Analytics
☐ IT Support
☐ Networking
☐ Cybersecurity
☐ Cloud Computing

☐ AI / Machine Learning

☐ Other: _____

Professional Statement:
"I am a BSIT graduate with skills in _____, _____, and _____. I am interested in pursuing a career as a _____."

STEP 2 — Conduct a Personal Skills Inventory

List everything you have learned during your BSIT program. Indicate your level honestly and identify evidence that proves the skill.

STEP 3 — Create Your Professional Email

Create a professional email account. Recommended format: firstname.lastname@gmail.com. Use it consistently for your resume, GitHub, LinkedIn, portfolio, and job applications.

STEP 4 — Create Your GitHub Account

Create a professional GitHub account. Learn to create repositories, upload files, clone, commit, push, pull, create branches, and write README files. GitHub will serve as your technical evidence repository.

STEP 5 — Create Your LinkedIn Profile

Create or update your professional LinkedIn profile. Include a professional photograph, name, headline, About section, education, technical skills, projects, certifications, GitHub link, and portfolio URL.

STEP 6 — Create Your Professional Resume

Create a one-page IT resume containing your name/professional title, contact information, professional summary, technical skills, education, projects, certifications, achievements, and optional references.

STEP 7 — Create Your Portfolio Website

The website itself is part of your assessment. Minimum technologies: HTML, CSS, JavaScript, and Bootstrap. You may additionally use React, Vue, PHP, Python Flask, Node.js, or other appropriate technologies.

STEP 8 — Create the HOME PAGE

The home page should immediately communicate who you are, what you do, what skills you have, what projects you have built, and how employers can contact you. Include clear buttons such as VIEW MY PROJECTS and DOWNLOAD RESUME.

STEP 9 — Create the ABOUT ME Page

Write approximately 150–250 words discussing your BSIT background, interests, technical strengths, career goals, professional values, and the type of IT work that interests you.

STEP 10 — Create the TECHNICAL SKILLS Page

Organize your skills into categories such as Programming, Web Development, Database, Tools, Networking, Cybersecurity, Data Analysis, Cloud, and AI. Only claim skills that you can demonstrate.

STEP 11 — Develop PROJECT 1: Web/Application Development

Create a functional application such as a Student Management System, Inventory System, Appointment System, Library System, E-Commerce System, School Management System, Booking System, or Point-of-Sale System.

Minimum requirements:

- User interface
- Database
- CRUD operations
- Validation
- Search
- Authentication
- Error handling

STEP 12 — Document PROJECT 1

Create a GitHub repository for Project 1. The README should include project overview, problem statement, objectives, target users, technologies, features, system architecture,

database design, screenshots, installation, usage, testing, limitations, future improvements, and developer information.

STEP 13 — Develop PROJECT 2: Data / Database Project

Create a project demonstrating your ability to work with data. Possible examples include Computer Shop Sales Analysis, Student Performance Analysis, Inventory Analytics, Small Business Sales Dashboard, or School Enrollment Analysis.

Use at least Excel/Google Sheets and SQL or a database, together with charts and data interpretation. Advanced students may additionally use Python, Pandas, Matplotlib, or Power BI.

STEP 14 — Develop PROJECT 3: Specialization Project

Choose one IT specialization: Web Development, Mobile Development, Cybersecurity, Networking, Data Analytics, AI, Cloud, or IT Support. The purpose is to demonstrate the IT area you want to develop professionally.

STEP 15 — Deploy at Least One Project

At least one project should be accessible online. Your portfolio should contain a live demo, GitHub repository, source code, and documentation. Understand basic hosting, deployment, environment variables, configuration, and troubleshooting.

STEP 16 — Prepare Project Documentation

For your major project, prepare requirements, system analysis, system design, development documentation, testing documentation, and deployment documentation.

Include functional and non-functional requirements, problem statement, existing/proposed system, use cases, architecture, UML, ERD, interface design, technologies, test cases/results, bugs and corrections, hosting, configuration, and installation information.

STEP 17 — Create Your Problem-Solving Portfolio

Document at least five IT troubleshooting cases.

For each case, record:
Problem
Possible Causes
Diagnostic Procedure

Solution
Lesson Learned

Example: Computer cannot connect to Wi-Fi. Possible causes may include disabled adapter, driver problem, incorrect IP configuration, router problem, or DNS problem.

STEP 18 — Create a Cybersecurity Case

Prepare one basic cybersecurity case study. Example: SQL Injection Vulnerability.

Explain:
1. What is the vulnerability?
2. How can it happen?
3. What are the risks?
4. How can developers prevent it?
5. What security practices should be implemented?

Focus on defensive understanding and responsible security practices.

STEP 19 — Prepare Your AI Literacy Evidence

Document how you used AI tools during the project. Create an AI Usage Log containing date, task, AI assistance, and your verification.

Answer:
• What did AI help me accomplish?
• What did I personally verify?
• What did I learn from using AI?
• What limitations/errors did I discover in AI-generated answers?

The purpose is to demonstrate responsible AI use, not dependence on AI.

STEP 20 — Prepare Your Technical Interview

Study programming, database, web development, networking, cybersecurity, and software engineering fundamentals. Be prepared to explain concepts and demonstrate practical problem-solving.

STEP 21 — Conduct the Mock Interview

Participate in a simulated IT job interview. Prepare answers for questions such as:
1. Tell me about yourself.
2. Why did you choose BSIT?

3. What are your strongest IT skills?
4. What is your weakest skill?
5. Tell us about your major project.
6. What was your biggest technical problem?
7. How did you solve it?
8. How do you handle deadlines?
9. How do you work with a team?
10. Why should an organization consider hiring you?
11. What technology are you currently learning?
12. Where do you want to develop professionally?

STEP 22 — Prepare Your Project Defense

Your final defense should include:
1. Introduction — 1–2 minutes
2. Problem — 2 minutes
3. Objectives — 1 minute
4. Proposed Solution — 2 minutes
5. Technology — 2 minutes
6. System Demonstration — 5–10 minutes
7. Database / Architecture — 3 minutes
8. Problems Encountered — 2 minutes
9. Solutions — 2 minutes
10. Future Improvements — 1 minute
11. Questions and Answers — Panel

STEP 23 — Prepare Your Career Plan

Create a one-page IT Career Development Plan identifying your target position, target IT field, current skills, skills to improve, technologies to learn, certifications to consider, 6-month goal, 1-year goal, and 3-year professional goal.

STEP 24 — Write Your Final Reflection

Write 500–800 words answering:
• What did I know about IT before entering BSIT?
• What were the most important things I learned?
• Which project taught me the most and why?
• What technical difficulty challenged me most?

• How have my IT skills changed?
• What skills do I still need to improve?
• What kind of IT professional do I want to become?
• Why am I now ready to begin my career in IT?

STEP 25 — Final Portfolio Checklist

Before submission, verify that all professional identity, technical skill, project, problem-solving, and employability requirements have been completed.

STEP 26 — Final Submission Structure

Submit both an online portfolio and an offline archive. The online version should connect your portfolio website, GitHub, live projects, and LinkedIn. The offline archive should contain organized folders for your profile, resume, projects, documentation, troubleshooting, security case, AI log, interview, career plan, reflection, and presentation.

SKILLS INVENTORY TEMPLATE

Area	Skill	Level	Evidence / Project
Programming	Python	Beginner / Intermediate / Advanced	
Programming	Java / C++ / C#	Beginner / Intermediate / Advanced	
Web	HTML / CSS	Beginner / Intermediate / Advanced	
Web	JavaScript / Bootstrap	Beginner / Intermediate / Advanced	
Database	SQL / MySQL	Beginner / Intermediate / Advanced	
Database	ERD / Database Design	Beginner / Intermediate / Advanced	
Tools	Git / GitHub	Beginner / Intermediate / Advanced	
Networking	TCP/IP / DNS /	Beginner /	

	Troubleshooting	Intermediate / Advanced	
Cybersecurity	Security Fundamentals	Beginner / Intermediate / Advanced	
Data	Excel / Sheets / Data Analysis	Beginner / Intermediate / Advanced	
Cloud	Deployment / Hosting	Beginner / Intermediate / Advanced	
AI	AI Tools / Responsible AI Use	Beginner / Intermediate / Advanced	

PROJECT 1 README TEMPLATE

1. Project Title
2. Project Overview
3. Problem Statement
4. Objectives
5. Target Users
6. Technologies Used
7. System Features
8. System Architecture
9. Database Design
10. Screenshots
11. Installation
12. How to Use
13. Testing
14. Limitations
15. Future Improvements
16. Developer / Contributors

TROUBLESHOOTING CASE TEMPLATE

Case Number:

_____

Problem:

_____

Possible Causes:

_____

Diagnostic Procedure:

_____

Solution:

_____

Lesson Learned:

_____

AI USAGE LOG TEMPLATE

Date	**Task**	**AI Assistance**	**My Verification / Result**

MY IT CAREER DEVELOPMENT PLAN

Target Position:

_____

Target IT Field:

_____

My Current Skills:

_____

Skills I Need to Improve:

_____

Technologies I Plan to Learn:

_____

Certifications I May Pursue:

_____

My 6-Month Goal:

_____

My 1-Year Goal:

_____

My 3-Year Professional Goal:

_____

FINAL PORTFOLIO CHECKLIST

☐ Professional email

☐ Professional photograph

☐ About Me

☐ Career objective

☐ Resume

☐ LinkedIn

☐ GitHub

☐ Programming skills demonstrated

☐ Web development skills demonstrated

☐ Database skills demonstrated

☐ Networking skills demonstrated

☐ Cybersecurity skills demonstrated

☐ Data analysis skills demonstrated

☐ Cloud/deployment skills demonstrated

☐ AI literacy evidence

☐ Project 1 completed

☐ Project 2 completed

☐ Project 3 completed

☐ Source code uploaded

☐ GitHub repositories completed

☐ Project documentation completed

☐ Screenshots included

☐ Testing documented

☐ At least one project deployed

☐ Five troubleshooting cases

☐ Security case study

☐ AI usage log

☐ Mock interview completed

☐ Technical interview completed

☐ Project defense completed

☐ Career plan completed

☐ Final reflection completed

FINAL STUDENT OUTPUT

At graduation, each student should be able to give an employer one professional portfolio link containing:

- About Me
- Technical Skills
- Projects
- Database Project
- Data Analysis
- Cybersecurity
- Web Development
- Cloud / Deployment
- AI Project or AI Literacy Evidence
- Troubleshooting
- Resume
- GitHub
- LinkedIn
- Interview / Employability Evidence
- Career Plan
- Reflection

- Contact Information

The portfolio should demonstrate: “I learned this. I built this. Here is the evidence. Here is my source code. Here is the live system. Here is how I solved the problem. And I can explain how it works.”

RECOMMENDED FINAL EMPLOYABILITY ASSESSMENT

Component	**Points**
Mock Job Interview	20
Technical Interview	25
Project Defense	30
Professional Communication	15
Career Plan	10
TOTAL	100

***IMPORTANT PRINCIPLE**
NO EVIDENCE = NO SKILL CLAIM.

If you claim a skill such as “Advanced Python,” you should be able to demonstrate and explain it. Your portfolio must provide evidence for the technical competencies you claim.

STUDENT PROJECT INFORMATION

Student Name: _____

Student ID: _____

Program: Bachelor of Science in Information Technology (BSIT)

Section: _____

School: _____

Instructor / Adviser: _____

Academic Year: _____

Portfolio URL: _____

GitHub URL: _____

LinkedIn URL: _____

BSIT INDUSTRY-READY PORTFOLIO
STEP-BY-STEP GUIDE FOR STUDENTS
From Planning → Building → Documenting → Presenting → Becoming Industry-Ready

1. PREPARATION & PLANNING

1 Define Your IT Professional Identity
- Know yourself
- Choose target IT field
- Write your professional statement



2 Conduct a Personal Skills Inventory
- List all skills learned
- Rate your level
- Provide evidence



2. SET UP YOUR ACCOUNTS

3 Create a Professional Email
- Use a proper format
- firstname.lastname
- Use for all accounts



4 Create a GitHub Account
- Create a repository
- Learn basic Git commands
- Prepare for project uploads



5 Create a LinkedIn Profile
- Add photo and headline
- Write about yourself
- Add skills and projects



6 Create a Professional Resume
- One-page IT resume
- Highlight skills, projects
- Use a clean format

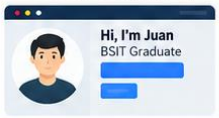


3. BUILD YOUR PORTFOLIO WEBSITE

7 Set Up Your Portfolio Website
- Use HTML, CSS, JavaScript, Bootstrap (or other tools)
- Plan your layout and menu



8 Create the Home Page
- Show your name, role, skills
- Add call-to-action buttons
- Make it visually appealing



9 Create the About Me Page
- Write 150–250 words
- Add your interests, strengths, and career goals



10 Create the Technical Skills Page
- Organize skills by category
- Use icons or progress bars

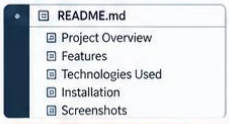


11 Develop Project 1 (Web/Application)
- Functional system
- CRUD, authentication
- Database integration



4. DEVELOP AND DOCUMENT YOUR PROJECTS

12 Document Project 1
- Create GitHub repository
- Use the README template
- Add screenshots and instructions



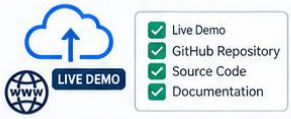
13 Develop Project 2 (Data/Database Project)
- Analyze real data
- Use Excel/SQL/Python/Power BI
- Create visualizations



14 Develop Project 3 (Specialization Project)
- Choose one IT specialization
- Build a related project
- Show your skills and interests



15 Deploy at Least One Project
- Host your project online
- Make it accessible
- Explain deployment process



16 Prepare Project Documentation
- Requirements
- System analysis and design
- Testing
- Deployment guide



5. PROBLEM-SOLVING & REAL-WORLD SKILLS

17 Create Your Problem-Solving Portfolio
- Document at least 5 troubleshooting cases
- Include solutions and lessons learned



18 Create a Cybersecurity Case
- Choose a security topic
- Explain the risk and prevention
- Focus on defensive understanding



19 Prepare Your AI Literacy Evidence
Document how you used AI tools
- Keep an AI usage log
- Explain your verification process



6. EMPLOYABILITY PREPARATION

20 Prepare for Technical Interview
- Review key IT concepts
- Practice technical questions
- Be ready to explain your projects



21 Conduct the Mock Interview
- Practice common questions
- Improve your communication



7. FINAL PREPARATION & SUBMISSION

22 Prepare Your Project Defense
- Follow the presentation structure
- Demonstrate your system
- Answer technical questions



23 Prepare Your Career Plan
- Set your target position
- List skills to improve
- Set 6-month, 1-year, 3-year goals

24 Write Your Final Reflection
- 500–800 words
- Discuss your growth, challenges, and future goals



25 Final Portfolio Checklist
- Check all requirements
- Ensure all links work
- Verify documentation

26 Submit Your Portfolio
- Online portfolio link
- GitHub repositories
- Supporting documents
- Prepare for final defense



FINAL STUDENT OUTPUT



RESULT: A COMPLETE, INDUSTRY-READY BSIT PORTFOLIO

- Showcases your skills
- Presents you professionally
- Demonstrates real projects
- Prepares you for employment
- Proves your problem-solving ability

PROPOSED I.T. SYSTEM PROJECT TO DEVELOP

#	Proposed Web-Based System Title	AI-Assisted Feature
1	AI-Assisted Student Academic Management System	AI Academic Assistant that answers academic/system-related questions and provides study suggestions based on student records.
2	AI-Assisted Student Performance Monitoring and Early-Warning System	AI analyzes grades and attendance and identifies students who may require academic intervention.
3	AI-Assisted Inventory and Point-of-Sale Management System	AI analyzes sales and inventory trends and provides stock/reorder recommendations.
4	AI-Assisted Library Management and Book Recommendation System	AI recommends books based on interests, subjects, borrowing history, or selected topics.
5	AI-Assisted Computer Repair Shop Management and Troubleshooting System	AI analyzes reported computer/device symptoms and provides possible troubleshooting procedures.
6	AI-Assisted Restaurant Ordering and Customer Recommendation System	AI recommends menu items based on customer preferences, previous orders, or selected categories.
7	AI-Assisted Clinic Appointment and Patient Records Management System	AI categorizes basic reported concerns and recommends an appropriate appointment/service category; it does not diagnose.
8	AI-Assisted Employee Payroll and Attendance Management System	AI analyzes attendance patterns and generates summaries or management insights from payroll and attendance data.
9	AI-Assisted Hotel	AI provides information about

#	Proposed Web-Based System Title	AI-Assisted Feature
	Reservation and Guest Support System	**rooms, facilities, policies, and assists guests in selecting suitable available rooms.**
10	**AI-Assisted Vehicle Rental Management and Recommendation System**	**AI recommends available vehicles based on passenger capacity, rental duration, budget, and user requirements.**
11	**AI-Assisted Personal Finance Management and Spending Analysis System**	**AI analyzes recorded transactions, identifies spending patterns, and provides budgeting suggestions.**
12	**AI-Assisted Learning Management and Personalized Study Recommendation System**	**AI analyzes quiz scores and learning progress and recommends appropriate learning resources or study activities.**
13	**AI-Assisted School Event and Registration Management System**	**AI recommends relevant school events based on student interests, course, or previous participation.**
14	**AI-Assisted Job and Internship Management System with Skill Matching**	**AI compares student skills and qualifications with stored job/internship requirements and identifies potential matches.**
15	**AI-Assisted E-Commerce Order and Product Recommendation System**	**AI recommends products based on customer interests, product categories, and previous transactions.**
16	**AI-Assisted IT Help Desk and Service Ticket Management System**	**AI classifies service tickets, identifies possible categories, and provides initial troubleshooting suggestions.**
17	**AI-Assisted Barangay Service Request Management and Classification System**	**AI automatically categorizes service requests and summarizes frequently reported community concerns.**

#	Proposed Web-Based System Title	AI-Assisted Feature
18	AI-Assisted Campus Facility Reservation Management System	AI recommends appropriate facilities based on activity type, capacity, schedule, and requirements.
19	AI-Assisted Sales and Customer Management System with AI Sales Insights	AI analyzes sales and customer records and provides summaries of sales patterns, customer activity, and frequently purchased products.
20	AI-Assisted Project and Task Management System with Intelligent Task Recommendations	AI analyzes project tasks and recommends priorities, task grouping, and possible next actions.